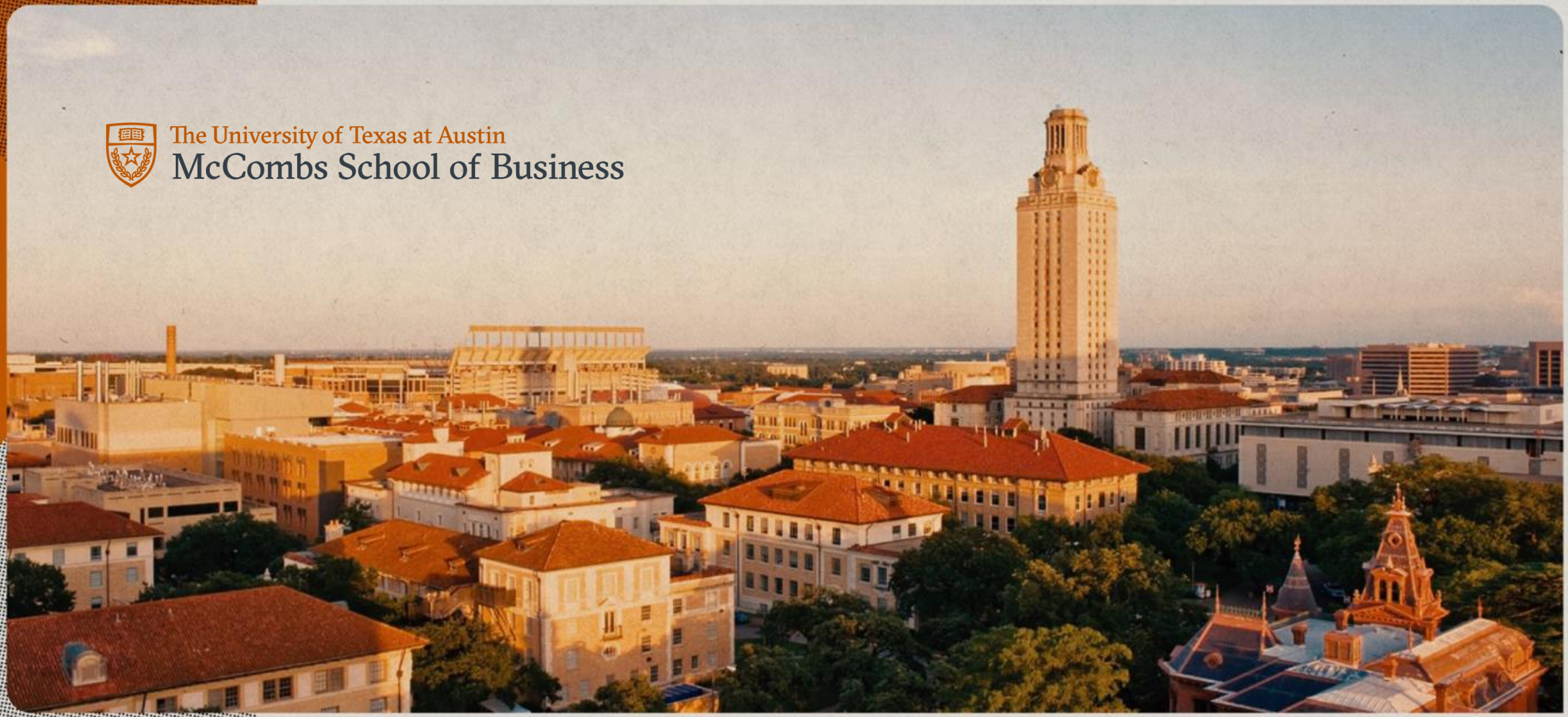


D4: Hierarchical agglomerative clustering

Learning Objectives

1. Understand hierarchical agglomerative (HA) approach to clustering
2. Implement HA clustering in R
3. Grow and interpret dendrograms
4. Compare and contrast HA with *K*-means

Major R commands

hclust

cutree

scale

rect.hclust

hclust sub-object: $height

aggregate

Hierarchical Agglomerative Cluster Analysis

$$X_{n \times p}$$

n rows of cases, records, observations, profiles

p columns of variables, fields, features

<u>Goal</u>: Discover row structure of X by grouping similar rows together and apart from dissimilar groups

How to Group?

Distance

Group cases that are close to each other in terms of Euclidean distance

Correlation

Group cases that are highly correlated with each other

Problems

Distance algorithm pays more attention to relatively small differences in large-valued variables (like income in $) than to relatively small differences in small-valued variables (like number of cars owned).

Correlation algorithm pays attention only to shape of profile, not to its elevation or scatter.

<See “Jane-Jim correlation and distance metrics for clustering.xlsx”>

Solutions

Distance algorithm: Standardize the analysis variables to treat all of them the same. This is the conventional solution.

But are all variables equally important? If multiple variables measure essentially the same feature (e.g., length, surface area, volume), then that feature gets more weight.

Correlation algorithm: Don't use it!

Distance

Euclidean Distance

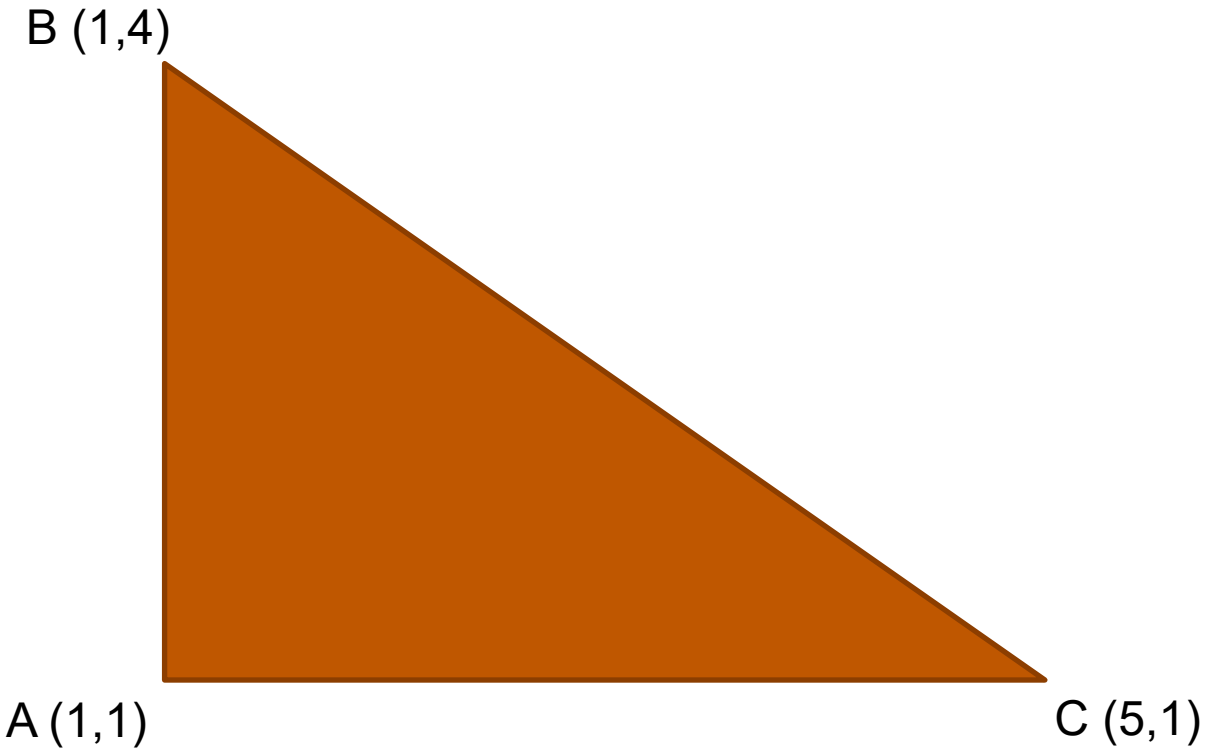
Pythagorean theorem

$BC^2 = AB^2 + AC^2$

$AB^2 = (1-1)^2+(4-1)^2= 9$

$AC^2 = (5-1)^2+(1-1)^2=16$

$BC^2 = (5-1)^2+(1-4)^2=25$



Distance Matrix

Distances of all pairs of cases (Squared Distances)

Example: A=(1,1), B=(1,4), C=(5,1)

$$\begin{array}{c c} & \begin{array}{ccc} A & B & C \end{array} \\ \begin{array}{c} A \\ B \\ C \end{array} & \begin{pmatrix} 0 & 9 & 16 \\ 9 & 0 & 25 \\ 16 & 25 & 0 \end{pmatrix} \end{array}$$

Hierarchical Agglomerative Algorithm

1. Begin with each case in its own singleton cluster. # clusters = n (cases)
2. Find two closest cases in distance matrix and merge them. #clusters = n-1
3. Recompute distance matrix, now (n-1)x(n-1). Need method to calculate distance between new 2-case cluster and other n-2 singleton clusters. Called the linkage problem. Some possibilities: min dist, max dist, average dist, centroid dist, Ward
4. Find two closest clusters in new distance matrix and merge them. #clusters = n-2
5. Recompute distance matrix, now (n-2)x(n-2), applying linkage method.
6. Continue until final two clusters merge into one supercluster with all n cases.
7. Decide where to stop the agglomeration.

Where to Stop the Agglomeration?

Dendrogram (Visual)	Rsquare (Quantitative)
Cut tree as close to the leaves as possible without making too many branches	Choose a hierarchy level as close to the end as possible with still high R-square

Balance explanatory power with interpretability.

hclust Demo

R program-CA stocks.R

Uses *Stock data.xlsx*

Illustrates use of *hclust* in R

Your turn

Use *hclust* with *ward.D2* method to hierarchically cluster *AustinApartmentRent.xls* using all variables except Rent. Use standardized variables.

What is the R-square for the 4-cluster solution?

Can you interpret the 4-cluster solution in original scale?

Do you prefer a clustering solution with a different number of clusters?